These programs are now also capable of extrapolating whether materials used to fabricate the 3D models will be functional or not. In earlier versions, designs made in the program were only true for geometric accuracy, but can now create true real world simulations that show how materials and complex shape geometries will behave in response to real world physics. The software's also have the ability to allow users to make their designs more efficient using lesser components which saves commercial industrial manufacturers on time and resources. The same benefit helps 3D printers by taking into account the additive printing system and compensating for that workflow.

## Surface Modeling CAD

As we know, CAD programs use vast x-y-z coordinate data sets to create a virtual object model. But using only the basic system isn't sufficient when dealing with objects like character shapes or complex molecular design. Surface modeling allows CAD programs to virtually enclose a shape or "wrap" a shape in a digital net or "mesh" giving it definition. This surface mesh is composed of numerous regular polygon geometries and is also known as polygonal modeling. Some people may be aware of this technique being used in 3D animation or game design where polygonal modeling is used to create lifelike looking characters and environments.

Unique and useful object shapes can be created in CAD by tweaking basic design shapes.

The polygonal modeling allows the program to correlate the coordinate sets along a virtual grid that corresponds to the mesh. All the individual coordinate data points are stored within the mesh surface model and allow the designer to accurately render the shapes. The default shape used for the mesh is typically triangular due to its efficient processing by computers but the level of detailing can range depending on the power of the processor. But it is important to note that these triangular shapes making up the mesh are themselves calculated on a planer surface, not a curved, much like how we look at maps as planar two dimensional surfaces and not the curved globe of which they are a part.

The great advantage of surface modeling programs is that it allows the program to replicate physical reality more accurately. The detailing